

Minor Project Report

On

"AI-Powered Voice Receptionist for Barber Shop Appointment Booking"

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for the 4th Semester Diploma in Computer Science & Engineering

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Certificate

This is to certify that the work in this project report entitled "AI-Powered Voice Receptionist for Barber Shop Appointment Booking" by Aritra Jana, Seikh Imaad Aktar, Tushar Kanta Panda, Sagar Mandal, and Maaz Manzar has been carried out under my supervision in partial fulfilment of the requirements for the Diploma in Computer Science & Engineering during the session Summer 2025-26 of C.V. Raman Polytechnic, and this work is the original work of the above students.

Guide – Ipsita Ankita Hota



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Abstract

This project presents an AI-powered voice receptionist system designed to automate appointment booking for a barber shop. The system integrates Vapi.ai for natural language voice interaction and n8n for backend workflow automation. The AI assistant, named Riley, is hosted on the Vapi platform and handles the complete booking lifecycle — checking slot availability by calling a POST webhook on the n8n server, collecting the customer's name and phone number, confirming the appointment, and notifying the shop owner via Telegram. The workflow reads from and writes to a shared Google Sheet which acts as the real-time booking database. This project demonstrates practical integration of AI voice agents, webhook-based automation, and notification pipelines in a real-world small-business context, using entirely free and open-source tools.

Chapter 1

Introduction

In the traditional barber shop model, appointment bookings are managed manually — either in person or over phone calls — which is time-consuming and prone to errors such as double bookings or missed appointments. This project addresses that problem by building an AI-powered voice receptionist system that automates the entire booking process using modern AI and workflow automation tools.

When a customer interacts with the system, they are greeted by Riley, an AI voice assistant configured on Vapi.ai. Riley conducts a natural conversation to identify the desired service, preferred date and time, and the customer's contact details. It then communicates with the n8n automation backend via custom webhook tools to check availability in a Google Sheet and confirm the booking — all without any human involvement from the shop side.

Technology Stack

- Vapi.ai — Cloud-based voice AI platform for building real-time voice agents with custom tool integrations
- n8n — Open-source workflow automation tool that serves as the backend logic engine, handling all webhook requests
- Google Sheets — Acts as a lightweight real-time database for storing bookings with columns: name, date, service, phone, start_time, end_time
- ngrok — Tunnelling tool that exposes the local n8n instance to the internet via a public HTTPS URL so Vapi can reach the webhooks
- Telegram Bot API — Used for both owner notifications (new booking alert) and customer confirmations (booking receipt)

The project follows a two-path architecture: Path 1 handles availability checking, and Path 2 handles booking confirmation. Both paths are triggered by Vapi tool calls and respond back to Vapi with structured JSON results.

Chapter 2

Objective of the Project

- To build a functional AI voice assistant (Riley) capable of handling barber shop appointment bookings through natural language voice conversation.
- To automate slot availability checking by querying a Google Sheet database in real time via an n8n webhook endpoint triggered by Vapi's checkAvailability tool.
- To ensure accurate slot booking by writing confirmed appointment data back to the Google Sheet with computed start and end times via the finalizeBooking tool.
- To implement a notification system that alerts the shop owner and confirms the booking to the customer using Telegram Bot API.
- To demonstrate integration of a voice AI agent (Vapi), a workflow automation engine (n8n), and a cloud database (Google Sheets) into a single cohesive automated pipeline.
- To create a team-replicable deployment that every group member can run independently using their own Google credentials and ngrok tunnel, following the project user guide.

Chapter 3

Relevance and Significance

Small businesses such as barber shops, salons, and clinics form a large segment of local commerce in India. Most lack the resources to hire dedicated receptionists or invest in expensive CRM software. This project is directly relevant because it demonstrates how freely available tools and affordable AI services can be combined to solve a real operational problem with zero recurring infrastructure cost.

Academic Significance

This project applies knowledge from multiple subjects in the Diploma CSE curriculum. Computer Networks concepts are applied through HTTP POST webhook communication, REST API calls, and JSON data exchange. The n8n workflow logic reflects principles of process automation and scheduling covered in Operating Systems. Google Sheets API integration relates to database and IT systems coursework. Additionally, the project introduces students to emerging industry concepts such as AI voice agents, LLM-based assistants, and no-code workflow orchestration.

Industry Relevance

Voice AI receptionist systems are an active area of commercial development, with companies like Twilio, Google, and Amazon investing heavily in similar technologies. By building a working prototype using Vapi.ai — a platform used by real startups — this project provides hands-on exposure to industry-grade tools and workflows. Understanding webhook-based tool calling, token-based API authentication, and multi-node automation pipelines are directly marketable skills in the software industry.

Social Impact

Automating appointment management reduces idle time for shop owners, eliminates missed or double bookings, and provides instant confirmation to customers. This directly benefits small business owners in Bhubaneswar and similar Tier-2 cities where smartphone adoption is high but digital business tools remain underutilized.

Chapter 4

Methodology

The project was developed in four distinct phases, each building on the previous one. The architecture follows a two-path design based on the Logic Blueprint defined in the project user guide.

Phase 1: System Architecture Design

The system was designed with two core workflow paths. Path 1 (Checking Availability) is triggered when Vapi calls the checkAvailability tool, sending a POST request to the n8n webhook at /webhook/check-availability with a date parameter. Path 2 (Finalizing the Booking) is triggered when Vapi calls the finalizeBooking tool, sending a POST request to /webhook/sierra-booking-success with full booking details. The Google Sheet acts as the shared state between both paths.

Phase 2: n8n Workflow Construction

Two n8n workflows were built. In Path 1, the workflow extracts the date from the Vapi payload, filters the Google Sheet for existing bookings on that date, computes all 30-minute free slots between 09:00 and 18:00, and returns the first five available slots as a formatted string back to Vapi. In Path 2, the workflow extracts all booking fields (name, phone, date, slot, service), computes the end time from the start time, appends the booking to the Google Sheet, and simultaneously sends Telegram notifications to both the owner and the customer.

Phase 3: Vapi.ai Configuration

The Riley assistant was created on the Vapi dashboard using the Appointment Scheduler template. Two custom tools — checkAvailability (with date and time as string parameters) and finalizeBooking (with name, phone, date, time, service, total_price, total_duration as string parameters) — were configured with their respective ngrok webhook URLs. The detailed system prompt defines Riley's identity, conversational flow (three phases: check, result, booking), IST timezone calculation logic using the {{now}} variable, service pricing reference, and critical operational rules.

Phase 4: Notifications and Team Deployment

Telegram notifications were configured using @BotFather to create a bot token and @userinfobot to retrieve the Chat ID. The Notify-owner node sends a structured booking alert to the shop owner. The Notify-Clients node sends a confirmation message to the customer. Each team member replicated the full setup on their own machine by following the user guide: importing the workflow JSON, creating their own Google OAuth2 credential, linking their own Google Sheet, setting the timezone to Asia/Kolkata, and updating the ngrok URL in Vapi tool settings.

Google Sheet Database Structure

The Google Sheet used as the project database contains three tabs as specified in the user guide. The Bookings tab stores all appointment records with the following columns: Booking_ID, Name, Phone, Date, Service, Start_Time, End_Time, Status, Total_Price, and Total_Duration. The Services tab stores the service catalogue with Service_Name, Price (INR), and Duration (mins). The Customers tab stores customer records including Name, Phone, Total_Visits, Last_Visit_Date, Total_Spends, Loyalty_Points, Cancellation_Count, Cancellation_Ratio, and Notes. The n8n workflow reads from and writes to the Bookings tab exclusively. The Booking_ID is auto-generated as a timestamp-based unique identifier to prevent duplicate entries.

ngrok Tunnel Configuration

Since n8n runs locally on the developer's machine, it is not directly accessible from the internet. ngrok solves this by creating a secure tunnel that maps a public HTTPS URL to the local n8n port (default: 5678). Before starting the workflow, each team member must run 'ngrok config add-authtoken <your_token>' once to register their account, then start the tunnel. The generated URL (e.g., <https://your-name.ngrok-free.app>) is then pasted into the Server URL fields of both Vapi tools — checkAvailability and finalizeBooking. The webhook paths /webhook/check-availability and /webhook/sierra-booking-success are appended to the base URL accordingly.

Chapter 5

Analysis – Testing

The system was tested end-to-end using the Vapi Talk button in the assistant dashboard, simulating real customer voice interactions. The following test scenarios were executed based on the workflow logic described in the user guide.

Test Case 1 – Successful Booking

A customer requested a Standard Haircut on a future date at a free time slot. Riley called checkAvailability, received a list of available slots, confirmed the slot was free, collected the customer name and a 10-digit phone number, and immediately called finalizeBooking. The booking was written to the Google Sheet with the correct start_time and computed end_time. Both Telegram notifications were delivered successfully.

Test Case 2 – Slot Already Booked

A time slot that was already present in the Google Sheet for the given date was requested. The get-availability node correctly identified zero overlap-free slots for that exact time. Riley responded with 'I'm sorry, that time is already booked' and offered alternative available times from the returned list, confirming the conflict detection logic works correctly.

Test Case 3 – IST Timezone Accuracy

The system prompt uses the {{now}} variable (server UTC time) and instructs Riley to mentally add 5 hours 30 minutes to derive Indian Standard Time. When a customer said 'book for today at 2 PM,' Riley correctly resolved 'today' and '2 PM' based on IST, and never suggested a slot that had already passed in India — confirming the timezone handling logic functions as designed.

Test Case 4 – Multi-Detail Input

A customer provided service, date, and time all in a single sentence ('I want a beard trim tomorrow at 11 AM'). Riley's SMART RECOGNITION rule correctly acknowledged all three pieces of information simultaneously and immediately called checkAvailability without asking redundant questions — confirming the system prompt logic for skipping already-provided fields.

Test Case 5 – Database Verification

After each successful booking, the Google Sheet was manually inspected. All six columns (name, date, service, phone, start_time, end_time) were populated correctly. The start/end time JavaScript node correctly computed 30-minute windows for both :00 and :30 minute slots.

Chapter 6

Result & Discussion

The project successfully achieved all stated objectives. The complete system was deployed and verified by all group members on their individual machines following the user guide.

Key Results

- A fully functional AI voice receptionist named Riley was deployed on Vapi.ai, capable of conducting end-to-end barber shop booking conversations without any human intervention.
- The n8n workflows correctly handled availability checking (Path 1) and booking confirmation (Path 2), with data accurately written to the Google Sheet Bookings tab.
- The Telegram notification pipeline delivered owner alerts and customer confirmations reliably upon every successful booking.
- All five team members successfully replicated the setup independently by following the user guide steps — creating their own Google OAuth2 credentials, configuring their own Google Sheets link, and updating ngrok tunnel URLs in Vapi.

Discussion

One operational limitation is the dynamic nature of the ngrok free-tier URL, which generates a new public domain every time n8n restarts. This requires updating the server URL in both Vapi tool configurations after every session restart. A production deployment would resolve this by using a paid static ngrok domain or hosting n8n on a cloud server with a fixed URL.

The availability check window in the n8n code is hardcoded as 09:00 to 18:00 in 30-minute intervals. However, the Vapi system prompt defines shop hours as 9 AM to 9 PM. This discrepancy between the code and the assistant's spoken response is a known gap that should be synchronized in a future iteration.

Overall, the system performs reliably within its defined scope and demonstrates how an AI voice agent and a workflow automation tool can be integrated to replace a human receptionist for routine scheduling tasks.

Source Code

The following is the complete source code of the n8n automation workflow extracted from the project JSON file. The workflow contains two parallel pipelines: Path 1 (Availability Check) and Path 2 (Booking Confirmation). All JavaScript node code is reproduced verbatim from the workflow.

1. Webhook: check-availability (Path 1 – Availability Check)

```
// Trigger: POST /webhook/check-availability
// Vapi sends: { body: { message: { toolCalls: [{ function: { arguments: { date } } } ] } } }

// Node: set - Extract date from Vapi payload
date = $json.body.message.toolCalls[0].function.arguments.date
```

2. Node: get-database – Google Sheets Lookup

```
// Operation: Read rows from Google Sheet (Bookings tab)
// Filter: rows where column "date" == extracted date
// Sheet ID: 1nfWprQgE2i9RqwiMTfs4wXF3tsIIHAUkf_-1S25eG_w
```

3. Node: check booked – Extract Booked Slots (JavaScript)

```
const booked = [];
for (const item of items) {
  const start = item.json?.["start_time"];
  const end = item.json?.["end_time"];
  if (start && end) {
    booked.push({ start, end });
  }
}
return [{ json: { booked } }];
```

4. Node: get-availability – Calculate Free Slots (JavaScript)

```
function toMinutes(t) {
  const [h, m] = t.split(':').map(Number);
  return h * 60 + m;
}
function toTime(m) {
  const h = String(Math.floor(m / 60)).padStart(2, '0');
  const min = String(m % 60).padStart(2, '0');
  return `${h}:${min}`;
}
const OPEN = toMinutes("09:00");
const CLOSE = toMinutes("18:00");
const SLOT = 30;
const bookedData = $input.item.json.booked || [];
const booked = bookedData.map(b => ({
  start: toMinutes(b.start),
  end: toMinutes(b.end),
}));
const available = [];
for (let t = OPEN; t + SLOT <= CLOSE; t += SLOT) {
  const overlap = booked.some(
    b => t < b.end && (t + SLOT) > b.start
  );
}
```

```

    if (!overlap) {
      available.push({ start: toTime(t), end: toTime(t + SLOT) });
    }
  }
  return [{ json: { available } }];

```

5. Node: filter – Format Response String (JavaScript)

```

const inputData = $input.all();
const slots = inputData[0].json.available || [];
const times = slots.slice(0, 5).map(slot => slot.start);
const timeSentence = times.join(', ');
return { resultString: timeSentence || "no openings" };

```

6. Node: Respond to Webhook – Return Available Slots to Vapi

```

// HTTP Response JSON sent back to Vapi AI:
{
  "results": [
    {
      "toolCallId": "{{ $node['getslot'].json['body']['message'],
        ['toolCalls'][0]['id'] }}",
      "result": "I found these openings: {{ $json['resultString'] }}.",
    }
  ]
}

```

7. Webhook: sierra-booking-success (Path 2 – Finalize Booking)

```

// Trigger: POST /webhook/sierra-booking-success
// Vapi sends: name, phone, date, slot, service

// Node: set1 - Extract all booking fields
Name      = $json.body.message.toolCalls[0].function.arguments.name
Date      = $json.body.message.toolCalls[0].function.arguments.date
Phone     = $json.body.message.toolCalls[0].function.arguments.phone
Slot      = $json.body.message.toolCalls[0].function.arguments.slot
Service   = $json.body.message.toolCalls[0].function.arguments.service

```

8. Node: start/end time – Compute End Time (JavaScript)

```

const input      = $input.item.json;
const slot       = input.Slot;    // e.g. "11:00"
let startTime    = slot;
let endTime;
if (slot.includes(':')) {
  let [hours, minutes] = slot.split(':');
  if (minutes === '00') { endTime = `${hours}:30`; }
  else if (minutes === '30') { endTime = `${parseInt(hours)+1}:00`; }
  else { endTime = `${hours}:59`; }
}
return {
  name: input.Name, date: input.Date,
  service: input.Service, phone: input.Phone,
  start_time: startTime, end_time: endTime
};

```

9. Node: get-database-update – Append Booking to Google Sheet

```
// Operation: Append row to Google Sheet (Bookings tab)
// Columns mapped:
//   name       = {{ $json.name }}
//   date        = {{ $json.date }}
//   service     = {{ $json.service }}
//   phone       = {{ $json.phone }}
//   start_time  = {{ $json.start_time }}
//   end_time    = {{ $json.end_time }}
```

10. Node: Notify-owner – Telegram Notification to Shop Owner

```
// Telegram Bot sends message to Chat ID: 7427007352
// Message Template:
" New Appointment, Boss! 🛎 "
" Client:  {{ $json.Name }} "
" Service: {{ $json.Service }} "
" Date:    {{ $json.Date }} "
" Time:    {{ $json.Slot }} "
" Phone:   {{ $json.Phone }} "
```

11. Node: Notify-Clients – Telegram Confirmation to Customer

```
// Telegram Bot sends message to Chat ID: 7427007352
// Message Template:
" Yay {{ $json.Name }}, you're all booked in!"
" Service: {{ $json.Service }} "
" Date:    {{ $json.Date }} "
" Time:    {{ $json.Slot }} "
```

12. Vapi.ai System Prompt (Riley's Intelligence)

[Identity]
You are Riley, a professional, friendly, and efficient AI receptionist for Imaad's Barber Shop. Your role is to guide customers through booking their barber services, ensuring accuracy and a seamless, human-like experience.

[Style]
- Use a warm, inviting, conversational tone.
- Speak clearly, using simple and polite language.
- Maintain professionalism but include natural speech elements.

[TIMEZONE & DATE HANDLING]
- UTC BASELINE: {{now}} (current server time).
- IST: India is UTC+5:30. Add 5h 30m to get Mumbai/Kolkata time.
- Interpret "Today", "Tomorrow", "Day after" based on IST.
- Never suggest a time that has already passed in Mumbai time.

[CRITICAL RULES]

1. As soon as you have Service, Date, and Time, call checkAvailability.
2. Once you have name and 10-digit phone, call finalizeBooking immediately.
3. Never mention tool names (checkAvailability, finalizeBooking) to customer.
4. Only check one time slot at a time.
5. For finalizeBooking, send ONLY numbers for total_price and total_duration.

[Pricing & Services Reference]

Hours: 9:00 AM to 9:00 PM (Mon-Sun)

- Standard Haircut: 250 INR (30 mins)
- Beard Trim: 150 INR (15 mins)
- Hair + Beard Combo: 350 INR (50 mins)
- Face Massage: 200 INR (20 mins)
- Skin Fade / Buzz Cut: 300 INR (45 mins)
- Hair Color / Dye: 600 INR (60 mins)
- Head Massage (Oil): 150 INR (15 mins)

GitHub Repository

The complete project source code, n8n workflow JSON file, and project documentation are hosted on GitHub. The repository includes the workflow file, the user guide, and any supporting assets required to replicate the setup.

Repository URL:

<https://github.com/Aritnatic/AI-Barber-Shop-Receptionist>

Chapter 7

Conclusion and Future Scope

Conclusion

This project successfully demonstrates the development of an AI-powered voice receptionist using Vapi.ai and n8n automation, backed by a Google Sheets database and a Telegram notification pipeline. The system eliminates the need for manual appointment management, reduces booking errors, and provides instant confirmation to both the customer and the shop owner. The integration of voice AI, webhook-based tool calling, JavaScript-based slot computation, and multi-channel notifications into a single cohesive workflow showcases the practical power of modern low-code automation tools combined with AI voice technology.

The project provided hands-on experience with REST API design, OAuth2 authentication, JSON payload handling, AI prompt engineering for voice agents, and team-based software deployment — skills directly relevant to the software industry.

From a systems design perspective, the project also highlights the importance of separating concerns across layers: the AI layer (Vapi) handles natural language understanding and conversation management, the automation layer (n8n) handles business logic and data operations, and the storage layer (Google Sheets) handles persistence. This layered architecture mirrors industry-standard microservice design patterns and demonstrates how complex, multi-component systems can be built and maintained by a small team using modular, loosely coupled components connected via standardized HTTP APIs.

Future Scope

- Deploy n8n on a cloud platform (such as Railway or a VPS) to eliminate the ngrok dependency and enable 24/7 availability without session restarts.
- Synchronize the shop's operating hours between the n8n availability computation code and the Vapi system prompt to eliminate the current 9 AM–6 PM vs 9 AM–9 PM discrepancy.
- Integrate a real-time cancellation and rescheduling flow so customers can modify existing bookings through the same voice interface.
- Replace Google Sheets with a proper relational database such as PostgreSQL or Supabase for better query performance and concurrent write safety.
- Add an SMS notification channel using the SMS Gateway Android app alongside Telegram, ensuring customers without the Telegram app also receive confirmations.
- Extend the system to support multiple staff members, each mapped to an individual availability calendar, enabling the assistant to route bookings to specific barbers.

References

1. Vapi.ai Documentation. "Building Voice Assistants." Vapi, <https://docs.vapi.ai>
2. n8n Documentation. "Workflow Automation with n8n." n8n, <https://docs.n8n.io>
3. Google Sheets API. "Sheets API Overview." Google Developers, <https://developers.google.com/sheets/api>
4. Telegram Bot API Documentation. <https://core.telegram.org/bots/api>
5. ngrok Documentation. "Getting Started with ngrok." <https://ngrok.com/docs>
6. SMS Gateway Android App. "API Documentation." <https://sms-gate.app/api>